



Spotted wing drosophila

Exotic to Australia

Features: A tiny fly (2–3 mm) that attacks a range of soft-skinned fruit and reduces crop yield and quality through direct feeding damage and secondary infection of the fruit

Where it's from: Originally from southeast Asia; has spread to the USA, Hawaii, Japan, Italy, Spain and France

How it spreads: Importation of larvae-infested plant material, primarily fruit; short-distance local spread by adult flies

At risk: Berries, cherries, peaches, nectarines, apricots, plums, prunes and grapes; apples, pears and other fruit with thicker skins once fruit begins to rot

Australian Chief Plant Protection Officer, Dr Gabrielle Vivian-Smith, provides an overview of the risks posed by spotted wing drosophila.

Spotted wing drosophila is a small fly that attacks a wide range of fruit. It is not present in Australia.

It has a significant impact on fruit production as the larvae feed on maturing fruit, not just over-ripe or decaying fruit.

It prefers soft-skinned fruits such as berry, cherry, nectarine, peach, plum, currant and grape. But apples, pears and other fruits with thicker skins are also hosts when fruit begins to rot.

Spotted wing drosophila can cause heavy losses from larval feeding and secondary infections with some berry growers in the United States reporting 100% fruit loss.

The adult flies are yellow brown in colour with dark bands on their abdomen and red eyes.

The easiest way to identify them is by the small dark spot on the front edge near the tip of each forewing. This spot is only found on male flies. Otherwise, they look almost identical to the regular vinegar fly.

The females are larger and can be identified under a microscope by their serrated ovipositor which allows them to lay their eggs in healthy fruit.

Spotted wing drosophila may be spread over long distances by the transport of infested fruit.

If you see flies with the distinguishing spot, try to get a sample and photo and be sure to report it.

Keep Australia safe by sourcing plant material from clean accredited suppliers and checking your orchard regularly.

Keep it out

Overseas, the spotted winged drosophila (*Drosophila suzukii*) is a major horticultural pest of thin-skinned fruit, including berries (blueberry, raspberry, blackberry, boysenberry, strawberry), stone fruit (cherry, nectarine, peach, apricot, plum, prune) and grape. It also attacks apples, pears and other fruit with thicker skins once fruit begins to rot.

Females lay eggs in fruit, causing feeding damage as well as rotting. While most flies can only lay eggs in ripe or damaged unripe fruit, this species can attack unripe and undamaged fruit. As a result, it can severely reduce yield, with some berry growers in North America reporting up to 100 per cent crop losses.

Spotted wing drosophila can spread rapidly, as the larvae hide in developing or ripe fruit while it is transported long distances from farms to markets. The adults can only fly short distances.

If this exotic fly established in Australia, it would cause serious losses for growers of soft berries, stone fruits and grapes.

Importing goods

To keep spotted wing drosophila out of Australia, never ignore Australia's strict biosecurity rules.

Import shipments may need to be treated and certified, so before you import, check our [Biosecurity Import Conditions system \(BICON\)](#).

What to look for

Adult flies are tiny, just 2–3 mm long. They have prominent red eyes, and their bodies are yellow–brown with dark bands on the abdomen. Males have a small, dark spot near the tip of each wing. This

black spot distinguishes the spotted wing drosophila from other *Drosophila* species present in Australia.

Fruit damage consists of pinprick-sized holes where females have laid eggs, followed by softening, skin wrinkling and fruit collapse from feeding. Fruit will often rot.

Larvae are cream or white and about 3 mm long.



Where to look

Importers

Importation of infested fruit is the most likely way that spotted wing drosophila would enter Australia.

Growers and home gardeners

Look for:

- larvae (tiny white grubs) in undamaged fruit still on the tree
- pinprick-sized holes or scars on the fruit surface
- rotting fruit.

What to do

If you think you've found spotted wing drosophila flies or larvae:

- do not disturb the insect (this may be as simple as closing the doors on a shipping container, or sealing the infested fruit in a plastic container)
- collect a sample, if it is possible to do so without disturbing the flies or larvae.

Source: <https://www.agriculture.gov.au/biosecurity-trade/pests-diseases-weeds/plant/identify/spotted-wing-drosophila>